



宁夏大学
NINGXIA UNIVERSITY

科研RA自荐

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学院：信息与网络空间安全学院

日期：2026.09.XX

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NINGXIA UNIVERSITY

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下一阶段计划

研究问题

来源：导师张若楠国家自然科学基金项目、宁夏自然科学基金优秀青年项目 RNS (Retrieve and Segment) 开放词汇分割框架

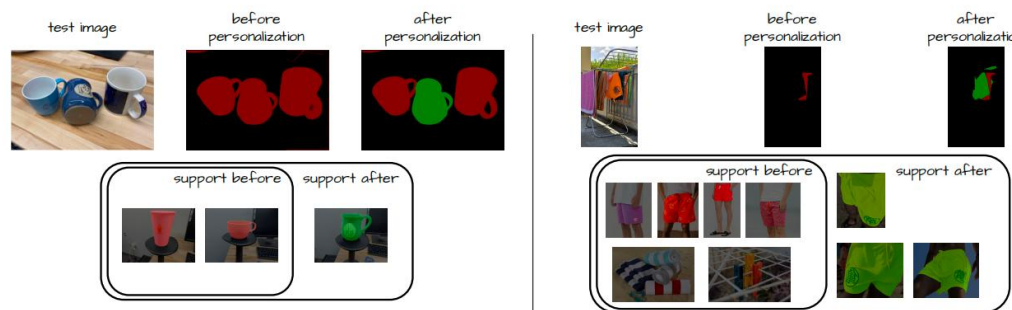
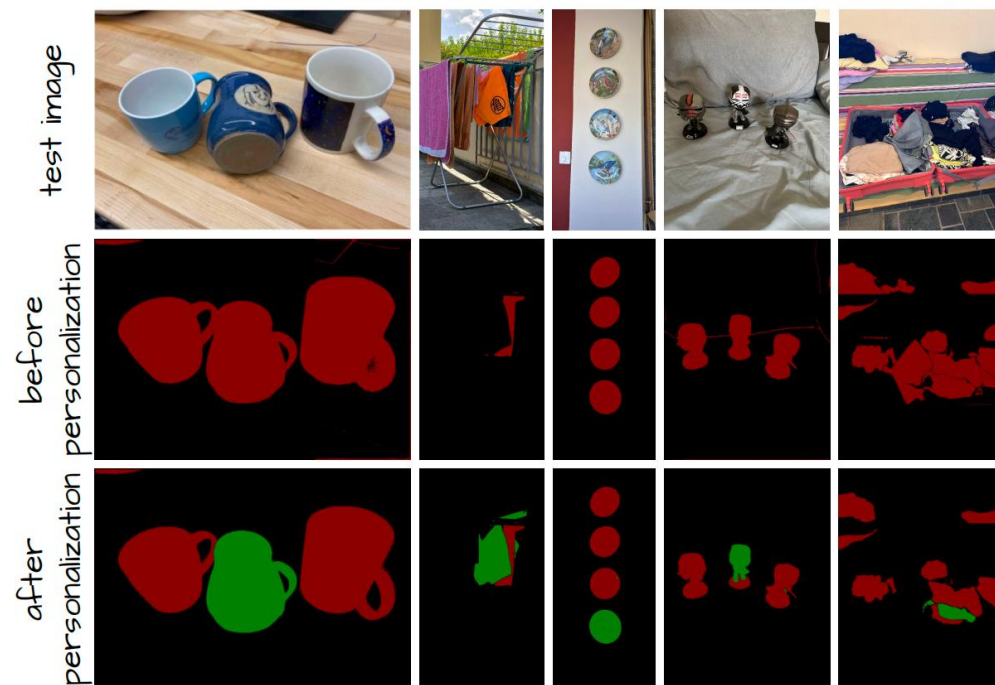
现有 RNS 在区域级检索后，会将重复命中的支持实例进行去重，只保留支持实例身份。这种做法能够减少冗余，但同时会丢失某个支持实例被当前 query 多次检索到的实例级多重性统计。

本研究关注：在不改变支持样本选择和类别级监督总量的前提下，如何保留并利用检索过程中产生的实例级命中信息，从而改善测试时视觉适配。

方法思路

在保持原 RNS 的支持构建、检索规则、融合方式、分类器和推断流程不变的基础上，首先统计当前 query 对各支持实例的重复命中次数，并在类别级视觉监督总权重保持不变的约束下，将该多重性信息转化为类内实例级监督分配。

类别级监督总量保持不变 + 检索诱导的类内实例分配



个人创新

我们提出 MP-RNS (Multiplicity-Preserving RNS), 保持原 RNS 的支持构建、检索规则、融合方式、分类器和推断流程不变的基础上, 首先统计当前 query 对各支持实例的重复命中次数, 并在类别级视觉监督总权重保持不变的约束下, 将该多重性信息转化为类内实例级监督分配, 计划投稿CVPR

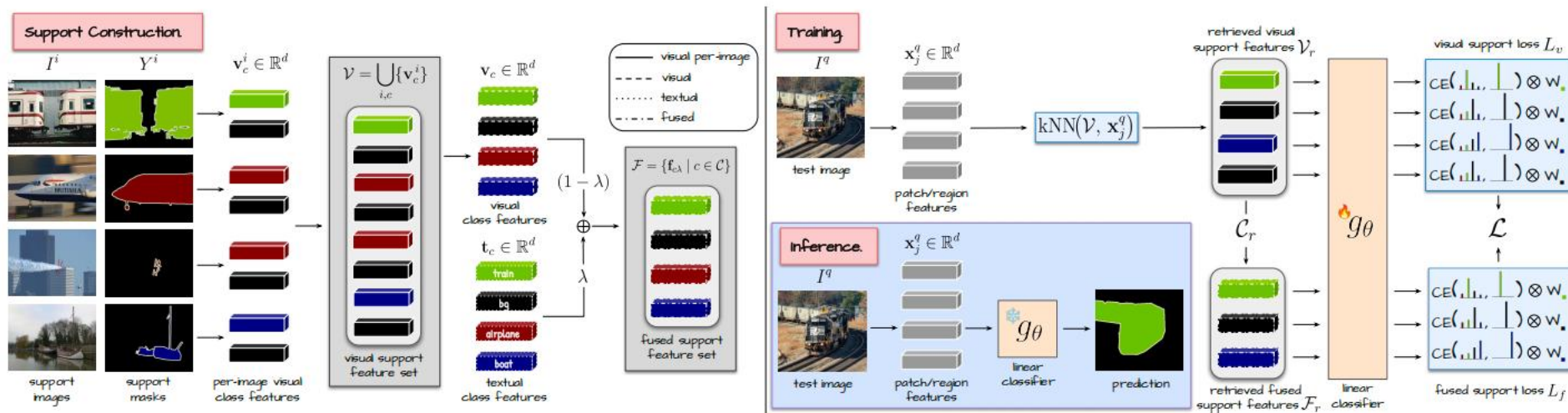


Figure 2. **Overview of RNS when full textual and visual support is available.** Having access to a set of pixel-level annotated images, per-image visual class features v_c^i are extracted. These features are then aggregated by class to form visual class features v_c , which are combined with textual class features t_c , through a mixing coefficient λ , to produce fused class features $f_{c\lambda}$. During test-time training, a test-image-relevant subset of visual support features and fused class features, along with their class labels, are used to train a lightweight linear classifier g_θ using cross-entropy loss. Each training sample is weighted with a class relevance weight w_c (e.g. w_{bg}). At inference, this classifier, trained per test image, is applied to patch-level features x_j^q to generate segmentation predictions. When SAM is available, patch-level features are replaced by region-level features x_r^q for improved accuracy.

1.2026PRCV参会 (2026.08哈尔滨)

作为**第一作者**参与论文 *Bayesian Reconstruction and Posterior-Projection Auditing of Expert-Crowd Elimination Mechanisms* 的研究与投稿。该研究针对专家评分公开而观众偏好不可直接观测的问题，采用贝叶斯重构、规则一致性筛选与 HMM 时序平滑，对专家-观众淘汰机制进行分析。论文以海报形式展示，**参加 PRCV 会议并参与现场学术交流，积累了论文展示、同行反馈和学术会议参会经验。**

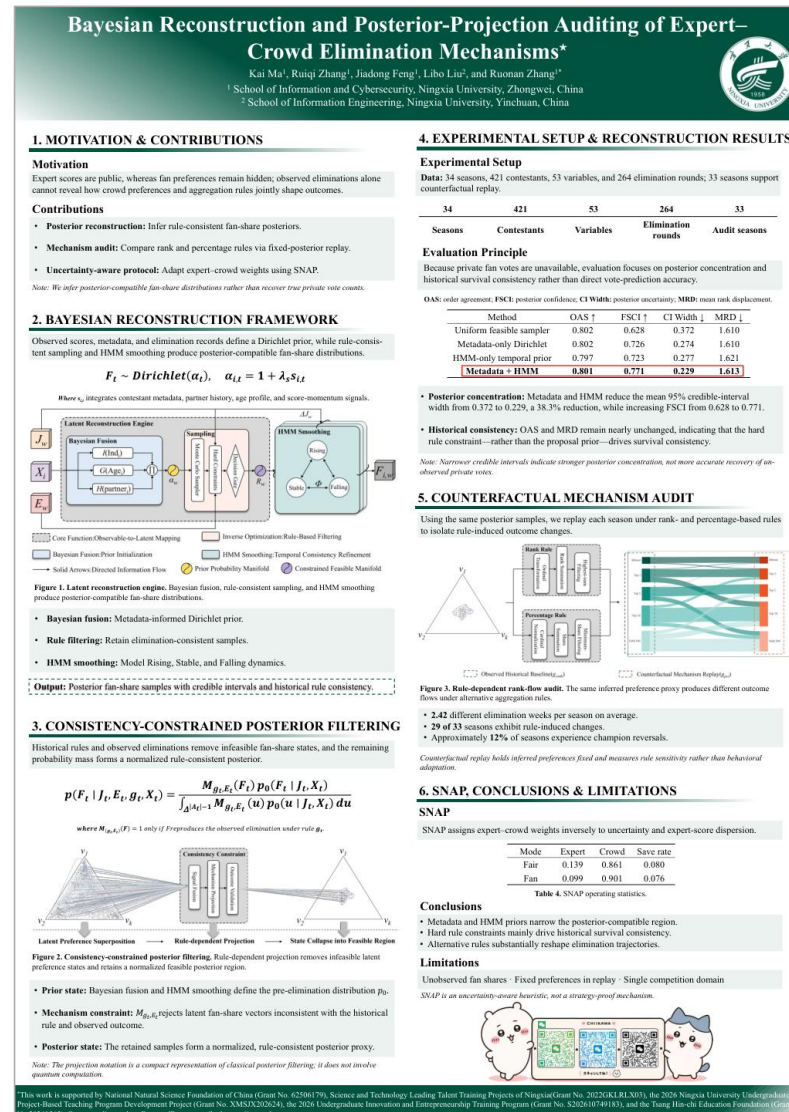
Bayesian Reconstruction and Posterior-Projection Auditing of Expert-Crowd Elimination Mechanisms*

Kai Ma¹, Ruiqi Zhang¹, Jiadong Feng¹, Libo Liu², and Ruonan Zhang^{1*}

¹ School of Information and Cybersecurity, Ningxia University

² School of Information Engineering, Ningxia University

Abstract. Expert-crowd elimination systems combine public expert scores with hidden audience preferences, making post-hoc mechanism analysis difficult. We formulate this setting as a human-centered pattern-analysis problem: infer feasible posterior fan shares, audit rule-induced changes, analyze expert-fan discrepancy, and test a dynamic scoring protocol. The method treats fan votes as constrained latent variables on a simplex and uses Bayesian fusion with Hidden Markov Model (HMM) smoothing to obtain posterior samples consistent with historical eliminations. A projection-form notation is used only as a compact description of classical rule-conditioned posterior filtering. Across 264 observed elimination rounds from 34 seasons, a four-way ablation shows that meta-



2.SMC (IEEE SMC 2026) 是系统科学、人工智能与人机协同等方向的国际学术会议 (CCF-C)，关注智能系统、计算智能、自然语言处理和人机交互等研究方向。

2026 年，以**第一作者**身份向 IEEE SMC 2026 投稿论文《Quantum-Inspired Dynamic Adaptive Framework for Textual Sentiment Analysis》，研究内容聚焦文本情感分析中的语义歧义建模与量子启发动态自适应干涉机制。目前论文已被 IEEE SMC 2026 录用。



IEEE SMC 2026
Bellevue, WA USA



IEEE
SMC
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IEEE

Quantum-Inspired Dynamic Adaptive Framework for Textual Sentiment Analysis

Kai Ma, Libo Liu, and Ruonan Zhang*

Abstract—Modeling with quantum theory shows promise in NLP, yet most rely on fixed interference terms and lack dynamic adaptive mechanisms, limiting their ability to handle varying sentence-level semantic ambiguity. To this end, we propose QDA, an end-to-end Quantum-inspired Dynamic Adaptive framework for textual sentiment analysis. It is the first to incorporate dynamic adaptive interference, which enables the effective resolution of semantic ambiguity via three dedicated modules. First, Quantum-inspired Embedding integrates pre-trained models with quantum encoding to capture rich semantic-syntactic information. Second, Quantum-inspired Dynamic Adaptive Interference adjusts interference strength according to sentence-level ambiguity, enabling flexible quantum feature fusion. Third, Quantum-inspired Sentiment Measurement performs measurements on quantum fused features to predict accurate and interpretable results. Finally, experiments on five benchmarks show that our method achieves SOTA performance with 94.00% average accuracy, outperforming existing methods. Additional analyses also support the demonstrated effectiveness and interpretability of the proposed method.

Index Terms—Sentiment Analysis, Quantum Theory, Quantum Interference, Quantum Measurement

I. INTRODUCTION

Textual sentiment analysis, the task of identifying subjective opinions in text, has become a cornerstone of modern Natural Language Processing (NLP). Traditional approaches have evolved from lexicon-based methods to deep learning models, e.g., CNNs, RNNs, LSTMs, and GRU series [1]. More recently, pre-trained language models have revolutionized NLP, evolving from static embeddings to contextualized models such as ELMo [2] and BERT [3].

However, classical models typically represent words as static, real-valued vectors. This representation often struggles to capture the probabilistic nature and inherent ambiguity of human language, such as polysemy and subtle emotional shifts [4], [5]. To address this, quantum-inspired modeling has emerged as a promising paradigm. By mapping semantic features into a complex-valued Hilbert space, these models

utilize quantum concepts to model linguistic ambiguity, such as superposition, interference. Quantum probability has been successfully adapted for text representation via complex word embeddings [6], density matrices [7], [8], and complex-valued networks [9]. Pioneer works like QLM [10] and Neural Network QLM [11] utilize density matrices to model term dependencies. Recent variational quantum circuits [12], [13] and quantum-inspired sentiment models [14]–[17] further demonstrate the potential of parameterized quantum operations for feature learning.

Despite these advances, existing quantum-inspired textual sentiment analysis methods still suffer from simplistic feature fusion strategies, e.g., QPFE-ERNIE [15] employs concatenation-based fusion where quantum-inspired and pre-trained features are simply concatenated before classification. Analogous to a fixed double-slit experiment [18], [19], such models apply a static fusion mechanism across all samples, ignoring the varying complexity of sentiment expressions. As illustrated in Fig. 1, we argue that the feature interaction mechanism should be **dynamic and adaptive**. E.g., the sentiment of explicit sentences is deterministic and behaves akin to a classical particle with minimal interference, where fixed interference [15] may introduce unnecessary noise. In contrast, the sentiment of ambiguous sentences resides in a semantic “superposition” state, necessitating stronger interference to capture complex nonlinear interactions. These distinct cases call for a dynamic and adaptive interference intensity measurement; otherwise, simpler cases may be overcomplicated, which could negatively impact overall performance.

This design draws on quantum cognition theory [20], which formally models ambiguity-driven judgments via interference dynamics, distinguishing our approach from purely metaphorical uses of quantum formalism.

Therefore, we propose an **end-to-end Quantum-Inspired Dynamic Adaptive framework** for textual sentiment analysis that integrates three key modules for precisely measuring the sentiment of sentences. First, we design Quantum-inspired Embedding (QE) to model semantic ambiguity by extending real-valued embeddings to the complex ones. Second, we devise Quantum-Inspired Dynamic Adaptive Interference (QDAI) to address the noise problem in fixed interference by adaptively adjusting intensity based on sentence complexity. Third, we apply Quantum-Inspired Sentiment Measurement (QSM) to produce accurate sentiment predictions from quantum-fused features. Extensive experiments on five benchmarks demonstrate state-of-the-art performance (94.00% average accuracy), with analyses confirming effective

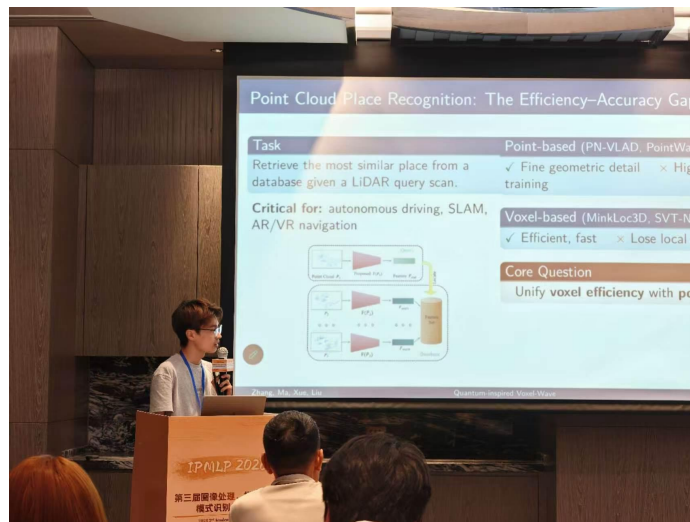
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This work is supported by the National Natural Science Foundation of China (62506179), Science and Technology Leading Talent Training Projects of Ningxia (2022GKLRXLX05), 2026 Ningxia University Undergraduate Project-Based Teaching Program Development Project (XM-SJX202624), and the Innovation and Entrepreneurship Training Program for College Students of Ningxia University (S202610749183 and S202610749184). We would like to thank the editors and reviewers for their time and constructive feedback, as well as everyone who assisted us during the course of this work. (Corresponding author: Ruonan Zhang)

3.2026IPMLP参会 (2026.05银川)

2026 年，以共同**第一作者**身份向 IPMLP 2026 投稿论文《Quantum-inspired Voxel-Wave: Bridging Efficiency-Accuracy Gaps in Point Cloud Place Recognition》，研究内容聚焦点云地点识别中的效率—精度权衡问题，**提出量子启发的 Voxel-Wave 框架**，通过结合体素表示与波函数建模，在**降低计算开销的同时提升地点识别性能**。目前论文已被 IPMLP 2026 录用。IPMLP (IPMLP 2026) 是面向图像处理、机器学习与模式识别领域的国际学术会议，关注计算机视觉、图像处理、机器学习和模式识别等研究方向。



Quantum-inspired Voxel-Wave: Bridging Efficiency-Accuracy Gaps in Point Cloud Place Recognition

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Abstract

Point cloud place recognition faces a fundamental efficiency-accuracy trade-off: point-based methods preserve fine-grained geometry but incur high computational costs, whereas voxel-based approaches improve efficiency at the expense of geometric fidelity. Inspired by quantum mechanics' wave-particle duality, we propose a Quantum-inspired Voxel-Wave Framework to bridge this gap by unifying the strengths of both paradigms via two synergistic modules. Our key insight is to treat point clouds as probability distributions governed by wave functions, enabling simultaneous encoding of structural detail and spatial relationships with controllable computational budgets. Specifically, Quantum Voxel-Wave Representation (QVWR) leverages spherical coordinate transformation and wave-function modeling to mitigate sparsity while retaining rich geometry, and Quantum Voxel-Wave Encoding (QVWE) employs a memory-efficient positional encoding and generalized mean pooling to enhance local feature awareness without heavy overhead. Extensive experiments on benchmark datasets demonstrate that our method achieves SOTA accuracy with significantly improved efficiency, effectively bridging the efficiency-accuracy gap. This work highlights the promise of quantum-inspired paradigms for advancing 3D scene understanding.

CCS Concepts

Computing methodologies → Scene understanding.

Keywords

point cloud, place recognition, quantum inspiration

ACM Reference Format:

Ruonan Zhang, Kai Ma, Feng Xue, and Libo Liu. 2026. Quantum-inspired Voxel-Wave: Bridging Efficiency-Accuracy Gaps in Point Cloud Place Recognition. In *Proceedings of The 3rd International Conference on Image Processing, Machine Learning, and Pattern Recognition (IPMLP 2026)*, July 10–12, 2026.

^{*}Ruonan Zhang and Kai Ma contributed equally to this work.

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IPMLP 2026, Yinchuan, China
2026. ACM ISBN 979-8-xxxx-xxxx-9/YY/AM
<https://doi.org/10.1145/xxxxxx.nnnnnn>

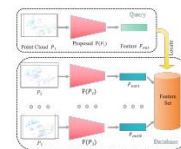


Figure 1: A query point cloud is matched against a point-cloud database to retrieve the most similar place.

Yinchuan, China. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/xxxxxx.nnnnnn>

1 Introduction

point cloud place recognition serves a critical function in robotics, self-driving, navigation, and AR/VR [2, 18, 26, 29]. Given a query scene (see Fig. 1), the framework retrieves the most similar match from a database by generating a compact scene representation via learning-based networks.

In the literature, point-based and voxel-based methods are widely adopted for this task. Point-based approaches excel at extracting fine-grained geometric features via point-level analysis, as exemplified by PN-VLAD [18], PointGem [21], LPD [17], PCAN [32], SONet [24], EPC-Net [7], PPT-Net [8], ERINet [22], Point-Wave [14], ComPoint [29], and POE-Net [15]. Despite their remarkable ability to preserve detail, these methods often incur substantial computational overhead and require extended training cycles. Voxel-based methods with sparse convolutions are generally more efficient, including MinkLoc3D [10], SparseARFM-SI [16], TransLoc3D [25], SVT-Net [4], MinkLoc++ [12], MinkLoc3Dv2 [11], PTC-Net [2], CASSPR [26], and QMM [28]. However, their sparse structures may still limit local feature learning and hinder further gains. Even emerging diffusion-style approaches [6, 13, 19, 20, 31, 34] continue to rely fundamentally on these underlying architectures. This landscape reveals a fundamental accuracy-efficiency trade-off, prompting a core research question: can we develop a simpler and more

02 学术成果:



4.在写的 CVPR 论文

提出 MP-RNS, 在原 RNS 检索与适配框架上, 保留 **query-conditioned retrieval multiplicity**, 并在类别总权重守恒约束下完成**类内实例监督重分配**, 使去重后的支持表示继续携带多重性信息参与训练, 同时**不增加可学习参数**

Method	$B = 1$	$B = 5$	$B = 10$
RNS (paper)	41.59	47.87	49.02
RNS w/o w_c	41.20	47.43	48.54
RNS w/o $w_c \Lambda = \{0.8\}$	36.40	46.55	48.38
RNS w/o text	34.11	45.71	48.00
MP-RNS (ours)*	42.93 +1.34	48.88 +1.01	49.90 +0.88

Method	Training set	Annot.	Backbones	VOC	City	ADE	C-59	Food	CUB	Avg.
ODISE [84]	COCO	118k	Stable Diffusion v1.3, CLIP (ViT-L/14)	84.6	—	29.9	57.3	—	—	—
OVSeg [†] [40]	COCO	118k	CLIP (ViT-L/14), Swin-B	—	—	29.6	55.7	16.4	14.0	—
SAN [†] [85]	COCO	118k	CLIP (ViT-L/14)	—	—	32.1	57.7	24.5	19.3	—
CAT-Seg [12]	COCO	118k	CLIP (ViT-L/14)	82.5	47.0*	37.9	63.3	33.3	22.9	47.8
LPOSS+ [67]	×	×	OpenCLIP (ViT-B/16), DINO (ViT-B/16)	62.4	37.9	22.3	38.6	26.1	12.0	33.2
CLIP-DINOiser [79]	×	×	OpenCLIP (ViT-B/16), DINO (ViT-B/16)	62.1	31.7	20.0	35.9	—	—	—
CorrCLIP [90]	×	×	OpenCLIP (ViT-H/14), DINO (ViT-B/8), SAM2.1 (Hiera-L)	76.4	49.9	28.8	47.9	—	—	—
MaskCLIP [97] + SAM	×	×	OpenCLIP (ViT-B/16), SAM2.1 (Hiera-L)	54.4	37.1	22.9	38.0	23.0	15.0	31.7
+ RNS $B = 1$	Domain	66	OpenCLIP (ViT-B/16), SAM2.1 (Hiera-L)	68.6	46.1	28.6	45.9	28.1	24.9	40.4
MP-RNS* $B = 1$	Domain	66	OpenCLIP (ViT-B/16), SAM2.1 (Hiera-L)	70.01	46.76	30.32	—	—	—	—
+ RNS $B = 20$	Domain	964	OpenCLIP (ViT-B/16), SAM2.1 (Hiera-L)	76.2	52.5	38.5	54.4	37.2	50.6	51.6
MP-RNS* $B = 20$	Domain	964	OpenCLIP (ViT-B/16), SAM2.1 (Hiera-L)	77.55	52.69	39.30	—	—	—	—
DINOv3.txt [65] + SAM	×	×	DINOv3 (ViT-L/16), SAM2.1 (Hiera-L)	31.3	39.3	27.7	36.3	27.2	5.8	27.9
+ RNS $B = 1$	Domain	66	DINOv3 (ViT-L/16), SAM2.1 (Hiera-L)	73.2	59.1	37.3	52.7	42.8	34.0	49.9
MP-RNS* $B = 1$	Domain	66	DINOv3 (ViT-L/16), SAM2.1 (Hiera-L)	—	—	—	—	—	—	—
+ RNS $B = 20$	Domain	964	DINOv3 (ViT-L/16), SAM2.1 (Hiera-L)	82.1	61.7	47.8	62.5	52.2	65.2	61.9
MP-RNS* $B = 20$	Domain	964	DINOv3 (ViT-L/16), SAM2.1 (Hiera-L)	—	—	—	—	—	—	—
InternImage [74]	Domain	—	InternImage-H	—	87.0	62.9	70.3	—	—	—
SETR [95]	Domain	—	SeTR-MLA	—	76.7	48.6	—	45.1	—	—
GFN [92]	Domain	—	ResNet-101	—	—	—	—	—	84.6	—
Mask2Former [10]	Domain	—	DINOv3 (7B)	90.4	86.7	63.0	—	—	—	—
Fully Supervised	Domain	20k	Best of Above	90.4	87.0	63.0	70.3	45.1	84.6	73.4

Table 4. **OVS vs. fully supervised segmentation.** *Fully Supervised*: best method picked per dataset. *: self-evaluated, †: from CAT-Seg. Mask2Former numbers from DINOv3 [65] paper. *Domain*: using annotations from each training set. *Annot*: the number of pixel-level annotated images that each method uses. For *Domain* we report the ADE annotations. RNS and competitor values are RNS (paper) unless marked, while MP-RNS values are MP-RNS (ours). Each MP-RNS row is positioned directly below the corresponding RNS row; this is not a paired comparison unless an RNS (reproduced) run shares the same seed/query/protocol metadata. ‘—’ denotes a result that is not yet validated.

目前这篇 MP-RNS 已经从idea 探索阶段进入了比较明确的 实验收尾, 论文叙事冻结, 等待正式写作的阶段。

整体上, 方法本身已经稳定, 不再需要继续找第二创新, 剩下的主要工作是把已有证据整理成论文。

03 项目实践：

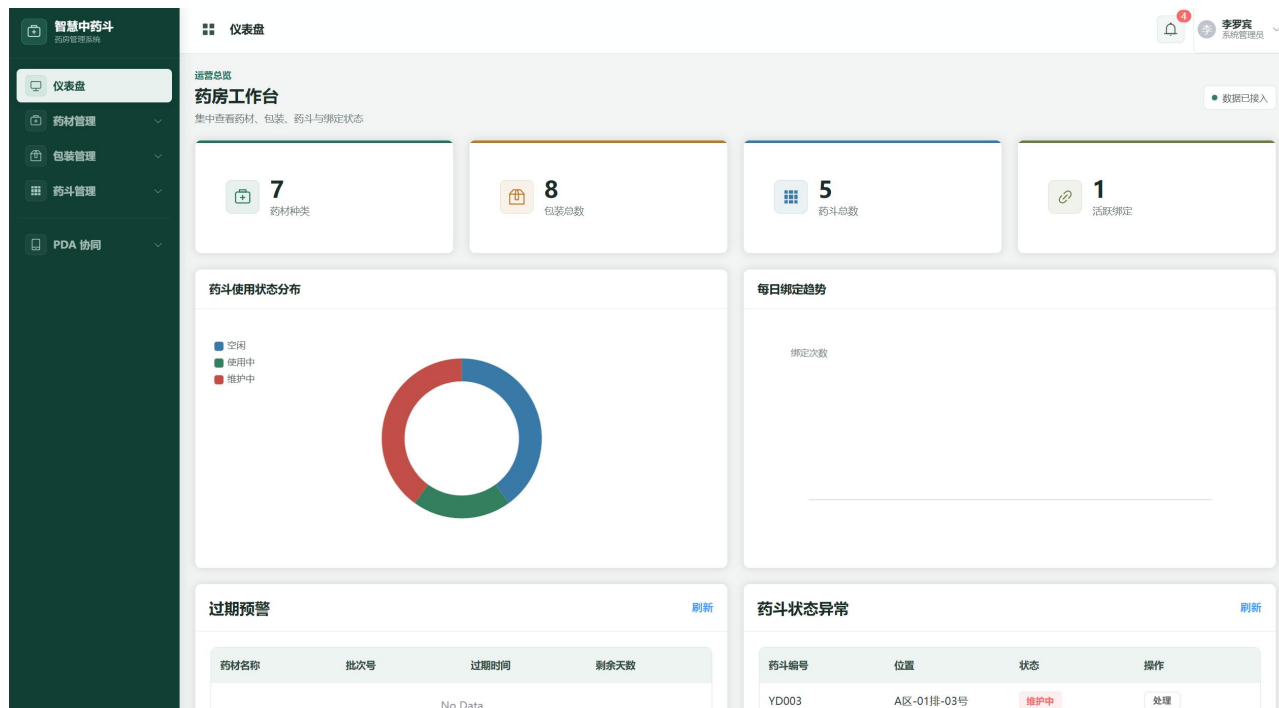


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1.作为2026**自治区级**大学生创新创业训练计划项目《基于 PDA 智能核验与动态绑定的智慧中药斗防差错系统》的**核心成员**



学院 | 自治区级 | 创新训练 | 基于PDA智能核验与动态绑定的智慧中药斗防差错系统 | 张睿琦



2.中国科学院神经科学研究所实习

2026年7月至10月，在**中国科学院神经科学研究所**类脑认知计算与脑机融合智能方向开展科研实习，参与科研项目的数据处理、模型分析与实验工作。**通过参与真实科研任务，进一步熟悉了科研项目的实验流程，积累了跨学科研究与团队协作经验。**

附件4（双面打印）

□新增 □续签

中国科学院脑科学与智能技术卓越创新中心 临时用工协议书

甲方（用人单位）：类脑认知计算与脑机融合智能

乙方（劳动者）：马凯

身份证号码：62082220060729381X

甲乙双方经协商，就乙方在甲方从事劳务工作、甲方支付劳务报酬等事项达成一致，订立本协议，共同遵守。

一、协议期限

自 2026 年 7 月 1 日起，至 2028 年 9 月 1 日止。

二、工作时间

非全日制，具体工作日及每个工作日的作息时间由甲方安排，工作地点上海市，需确保科研工作进度。甲方保障乙方的休息休假权。

三、工作内容及要求：乙方须完成或达到以下工作任务和要求，具体包括（但不限于）：

1. _____

四、劳务报酬

本协议履行期内，在乙方按时、按质、按量完成工作任务的前提下，甲方支付乙方每月（2000）元人民币的劳务报酬（税前），乙方劳务报酬可做浮动调整，如承担工作以外撰写项目申请书或者撰写项目指南或者执行多个项目的工作时，按研究组内规定需与组长交流，可三个月调整一次报酬。本报酬总额中已包含了乙方在本协议期限内为其自己购买必要的医疗保险和意外伤害保险的费用。

五、本协议的终止和解除

1. 本协议期限届满即告终止。



03 项目实践：



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3.其他相关奖项及项目实践：

- 美国大学生数学建模竞赛（MCM/ICM） **Outstanding Winner（最高奖）**；
- 智能货柜商品交易视频识别挑战赛优胜奖；
- 相关省级，国家级奖项若干；
- 中国大学生计算机设计大赛决赛入围奖；
- 三创赛校级一等奖；
- GitHub 开源项目：<https://github.com/Smile232323>

2026
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Ruiqi Zhang
With Faculty Advisor
Ruonan Zhang
or
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Outstanding Winner



-26-1555	宁夏大学	痕迹·轻量——基于空间融合痕迹的抗压深度伪造检测算法	王臻琪 陈宗华 时生榜 张鹏 王以升 李英 张芳艳	贰等奖
-26-1556	宁夏大学	故宫太和殿的建筑美学	潘玮杰 高铁瞳 朱睿 肖莉 李淼 李英	贰等奖
-26-1557	宁夏大学	智分宝—AI智能垃圾分类助手	郑国豪 朱睿 潘玮杰 高铁瞳 李英	贰等奖
-26-1558	宁夏大学	巧计英语：Harmony OS全场景高效单词学习平台	冯佳栋 赵文艳 耿莹 马丽 张馨悦 张若楠	贰等奖
-26-1559	宁夏大学	宁心——基于国产底座的大学生心理健康模型	马凯 张睿琦 耿莹 陈铭 王佳 张若楠	叁等奖
-26-1560	宁夏大学	智能药房药斗管理系统	张睿琦 马凯 白晶晶 杨超凡 汪江慧 张若楠	决赛入围奖



- 进一步**聚焦MP-RNS**研究，完成方法方案、基线复现、对比实验与消融分析，**推进论文撰写并准备CVPR会议投稿**
- 持续跟进**CVPR**论文的评审进展，根据反馈**完善实验材料和论文表达**
- **智慧中药斗大创项目**，将继续推进 PC 端与 PDA 端功能开发，完善扫码动态绑定、三级核验、数据追溯和异常预警等模块的联调测试，并**开展系统应用验证，整理项目文档和成果材料**

结合前期实习与学术会议经历，我将持续规范科研记录，提升独立开展研究、实验分析和成果表达的能力，按要求完成论文、实验数据和项目材料归档，为后续成果发表和项目结题做好准备



宁夏大学
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谢谢大家

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日期：2026.09.XX